

Лабораторная работа №1

Дисциплина: Администрирование сетевых подсистем

Бахи сиди али темассини

2026-02-16

1. Цель работы
2. Выполнение лабораторной работы
3. Выводы

Раздел 1

1. Цель работы

1.1 Цель работы

- Приобретение практических навыков установки Rocky Linux на виртуальную машину с использованием Vagrant

Раздел 2

2. Выполнение лабораторной работы

2.1 Конфигурационные файлы

- Созданы каталоги `C:\work\bahis\packer` и `C:\work\bahis\vagrant`
- Подготовлена структура для сборки box-файла

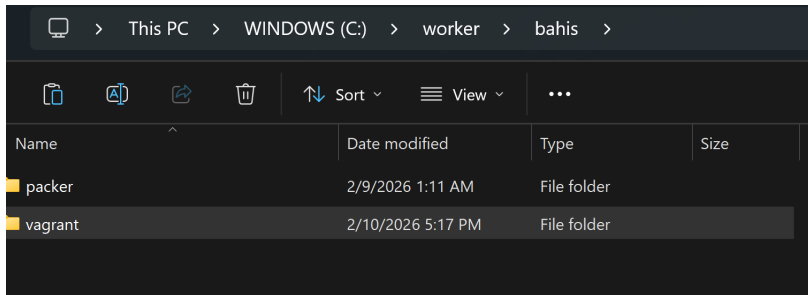


Рисунок 1: Структура рабочего каталога с подкаталогами packer и vagrant

- В каталоге `packer` размещены ISO-образ, `packer.exe`, HCL-файл и каталог `http`
- Структура соответствует требованиям проекта

 packer	11/18/2025 1:23 PM	Application	43,165 KB
 Rocky-9.4-x86_64-minimal	12/21/2025 8:03 PM	Disc Image File	1,786,752 ...
 vagrant-rocky.pkr.hcl	12/21/2025 9:52 PM	HCL File	4 KB

Рисунок 2: Содержимое каталога `packer` с ISO-образом и HCL-файлом

- Файл `vagrant-rocky.pkr.hcl` содержит блок `packer` и `required_plugins`
- Заданы переменные версии, диска, ISO, архитектуры и SSH
- В каталоге `http` размещён файл `ks.cfg`
- Определены параметры автоматической установки

```
packer {  
  required_plugins {  
    vagrant = {  
      source = "github.com/hashicorp/vagrant"  
      version = "~> 1"  
    }  
    virtualbox = {  
      version = "~> 1"  
      source = "github.com/hashicorp/virtualbox"  
    }  
  }  
}  
  
variable "artifact_description" {  
  type = string  
  default = "Rocky 9.4"  
}  
  
variable "artifact_version" {  
  type = string  
  default = "9.4"  
}  
  
variable "disk_size" {  
  type = string  
  default = "61440"  
}  
  
variable "iso_checksum" {
```



```
# System bootloader configuration
bootloader --append="no_timer_check console=tty0 console=ttyS0,115200n8 net.ifnames=0 biosdevname=0 elevator=noop" --location=mbr --timeout=1
# Clean the Master Boot Record
zerombr
# Partition clearing information
clearpart --all
# Reboot after installation
reboot
# Use text mode install
text
# Keyboard layouts
keyboard --vckeymap=us,ru --xlayouts='us,ru'
# System language
lang en_US.UTF-8
# Network information
network --bootproto=dhcp --device=link --activate
# System authorization information
authselect select sssd with-sudo with-mkhomedir --force
authselect apply-changes
# Root password
rootpw vagrant
user --name=vagrant --password=vagrant
firstboot --disable
# Do not configure the X Window System
#skipx
# System services
services --enabled="NetworkManager,sshd,chronyd"
# System timezone
timezone UTC --utc
user --name=vagrant --password=vagrant
# Disk partitioning information
part / --fstype="xfs" --size=10239
%post
# configure swap to a file
fallocate -l 2G /swapfile
chmod 600 /swapfile
mkswap /swapfile
echo "/swapfile none swap defaults 0 0" >> /etc/fstab
# sudo
echo "%vagrant ALL=(ALL) NOPASSWD: ALL" > /etc/sudoers.d/vagrant
chmod 0440 /etc/sudoers.d/vagrant
```

Рисунок 4: Файл ks.cfg с параметрами автоматической установки Rocky Linux

- В каталоге `vagrant` создан файл `Vagrantfile`
- Определена VM `server` с box `rocky9`
- Настроены IP `192.168.1.1` и параметры `VirtualBox`
- Создан каталог `provision` с подкаталогами `default`, `server`, `client`

```
# -*- mode: ruby -*-
# vi: set ft=ruby :
Vagrant.configure("2") do |config|

## Common configuration
config.vm.provision "common dummy",
type: "shell",
preserve_order: true,
path: "provision/default/01-dummy.sh"
config.vm.provision "common user",
type: "shell",
preserve_order: true,
path: "provision/default/01-user.sh"
config.vm.provision "common hostname",
type: "shell",
preserve_order: true,
run: "always",
path: "provision/default/01-hostname.sh"

## Server configuration
config.vm.define "server", autostart: false do |server|
server.vm.box = "rocky9"
server.vm.hostname = 'server'
server.vm.boot_timeout = 1440
server.ssh.insert_key = false
server.ssh.username = 'vagrant'
server.ssh.password = 'vagrant'
server.vm.network :private_network,
ip: "192.168.1.1",
__virtualbox__ :__interface__ => true
end
end
```

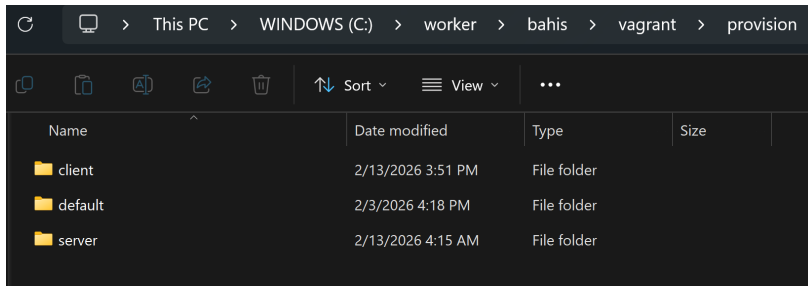


Рисунок 6: Структура каталога provision с подкаталогами default, server и client

- В каталогах размещены скрипты-заглушки 01-dummy.sh

```
#!/bin/bash  
  
echo "Provisioning script $0"
```

Рисунок 7: Скрипты-заглушки 01-dummy.sh в каталогах default, server и client

- В default добавлен 01-user.sh
- Создание пользователя bahis
- Добавление в группу wheel
- Настройка PS1

```
#!/bin/bash

echo "Provisioning script $0"

username=bahis
userpassword=123456

encpassword=`openssl passwd -1 ${userpassword}`

id -u $username
if [[ $? ]]
then
    adduser -G wheel -p ${encpassword} ${username}
    homedir=`getent passwd ${username} | cut -d: -f6`
    echo "export PS1='\u@\H \W]\\$ '" >> ${homedir}/.bashrc
fi
```

- В default добавлен 01-hostname.sh
- Установка FQDN вида *.bahis.net

```
#!/bin/bash

username=bahis

hostnamectl set-hostname "${HOSTNAME%.*}.${username}.net"
```

Рисунок 9: Скрипт 01-hostname.sh для установки доменного имени хоста

- В server размещён 02-forward.sh
- Включение IP-forwarding
- Настройка masquerading через firewalld

```
#!/bin/bash
echo "Provisioning script $0"
echo "Enable forwarding"
echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/90-forward.conf
sysctl -w net.ipv4.ip_forward=1
echo "Configure masquerading"
firewall-cmd --add-masquerade --permanent
firewall-cmd --reload
restorecon -vR /etc
```

Рисунок 10: Скрипт 02-forward.sh для включения IP-forwarding и masquerading

- В `client` размещён `01-routing.sh`
- Настройка шлюза `192.168.1.1`
- Конфигурация интерфейсов через `nmcli`

```
#!/bin/bash
echo "Provisioning script $0"
nmcli connection modify "eth1" ipv4.gateway "192.168.1.1"
nmcli connection up "eth1"
nmcli connection modify eth0 ipv4.never-default true
nmcli connection modify eth0 ipv6.never-default true
nmcli connection down eth0
nmcli connection up eth0
```

Рисунок 11: Скрипт `01-routing.sh` для настройки маршрутизации клиента

2.2 Развёртывание лабораторного стенда на ОС Windows

- Выполнены `packer.exe init` и `packer.exe build`
- Произведена автоматическая установка Rocky Linux
- Сформирован `box`-файл

```
C:\Windows\System32>cd C:\worker\bahis\packer

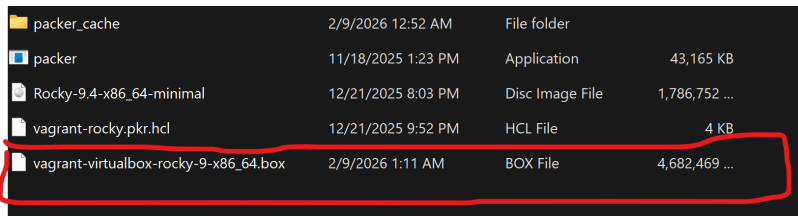
C:\worker\bahis\packer>packer.exe init vagrant-rocky.pkr.hcl

C:\worker\bahis\packer>packer.exe build vagrant-rocky.pkr.hcl
virtualbox-iso.virtualbox: output will be in this color.

==> virtualbox-iso.virtualbox: Retrieving Guest additions
==> virtualbox-iso.virtualbox: Trying C:\Program Files\Oracle\VirtualBox\VBBoxGuestAdditions.iso
==> virtualbox-iso.virtualbox: Trying file://C:/Program%20Files/Oracle/VirtualBox/VBoxGuestAdditions.iso
==> virtualbox-iso.virtualbox: file://C:/Program%20Files/Oracle/VirtualBox/VBoxGuestAdditions.iso => C:/Program Files,
acle/VirtualBox/VBoxGuestAdditions.iso
==> virtualbox-iso.virtualbox: Retrieving ISO
==> virtualbox-iso.virtualbox: Trying Rocky-9.4-x86_64-minimal.iso
==> virtualbox-iso.virtualbox: Trying Rocky-9.4-x86_64-minimal.iso?checksum=sha256%3Aee3ac97fdffab5865242194159990201
9c37535aece76824673105169c4a2
```

Рисунок 12: Выполнение команд `packer.exe init` и `packer.exe build`

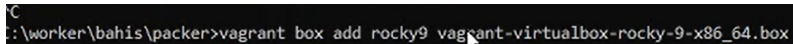
- Создан файл `vagrant-virtualbox-rocky-9-x86_64.box`
- Подтверждена корректная сборка образа



packer_cache	2/9/2026 12:52 AM	File folder	
packer	11/18/2025 1:23 PM	Application	43,165 KB
Rocky-9.4-x86_64-minimal	12/21/2025 8:03 PM	Disc Image File	1,786,752 ...
vagrant-rocky.pkr.hcl	12/21/2025 9:52 PM	HCL File	4 KB
vagrant-virtualbox-rocky-9-x86_64.box	2/9/2026 1:11 AM	BOX File	4,682,469 ...

Рисунок 13: Сформированный box-файл `vagrant-virtualbox-rocky-9-x86_64.box`

- Выполнена регистрация box через `vagrant box add rocky9`
- Образ добавлен под именем `rocky9`



```
C:\worker\bahis\packer>vagrant box add rocky9 vagrant-virtualbox-rocky-9-x86_64.box
```

Рисунок 14: Регистрация box-файла в Vagrant

- Запуск vagrant up server
- Инициализация сети и provisioning

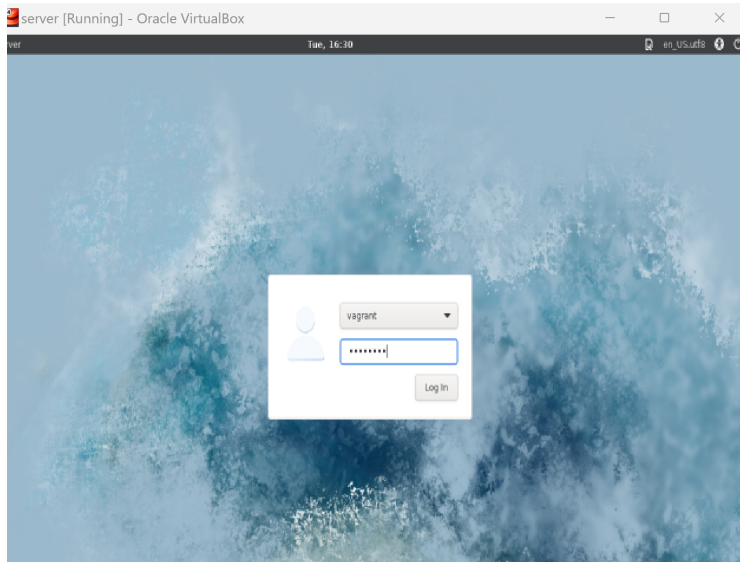
C:\> Administrator: Command Prompt

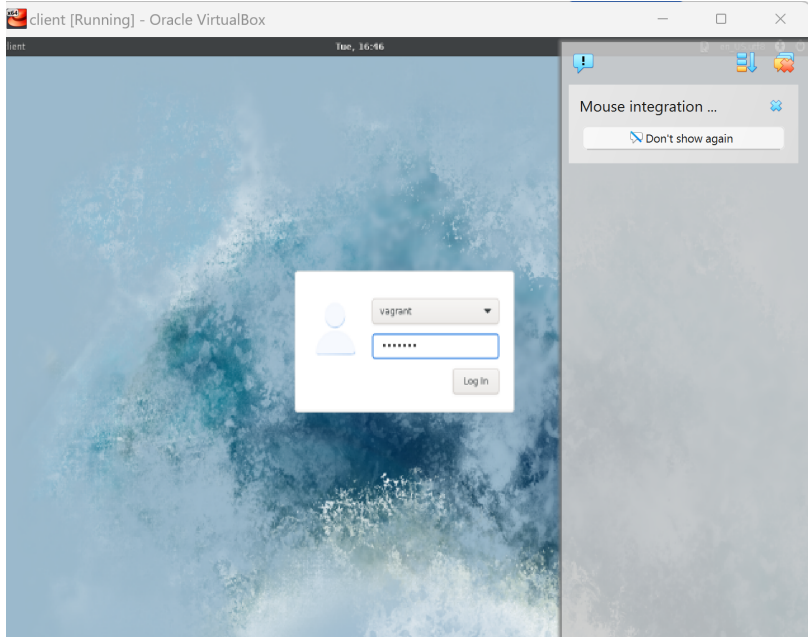
```
C:\worker\bahis\vagrant>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
=> server: You assigned a static IP ending in ".1" or ":1" to this machine.
=> server: This is very often used by the router and can cause the
=> server: network to not work properly. If the network doesn't work
=> server: properly, try changing this IP.
=> server: Cloning VM...
=> server: Matching MAC address for NAT networking...
=> server: You assigned a static IP ending in ".1" or ":1" to this machine.
=> server: This is very often used by the router and can cause the
=> server: network to not work properly. If the network doesn't work
=> server: properly, try changing this IP.
=> server: Setting the name of the VM: server
=> server: Clearing any previously set network interfaces...
=> server: Preparing network interfaces based on configuration...
    server: Adapter 1: nat
    server: Adapter 2: intnet
=> server: Forwarding ports...
    server: 22 (guest) => 2222 (host) (adapter 1)
=> server: Running 'pre-boot' VM customizations...
=> server: Booting VM...
=> server: Waiting for machine to boot. This may take a few minutes...
    server: SSH address: 127.0.0.1:2222
    server: SSH username: vagrant
    server: SSH auth method: password
=> server: Machine booted and ready!
```

- Запуск vagrant up client
- Настройка сетевых адаптеров
- Подключение по SSH

```
C:\worker\bahis\vagrant>
C:\worker\bahis\vagrant>vagrant up client
Bringing machine 'client' up with 'virtualbox' provider...
==> client: Cloning VM...
==> client: Matching MAC address for NAT networking...
==> client: Setting the name of the VM: client
==> client: Clearing any previously set network interfaces...
==> client: Preparing network interfaces based on configuration...
      client: Adapter 1: nat
      client: Adapter 2: intnet
==> client: Forwarding ports...
      client: 22 (guest) => 2222 (host) (adapter 1)
==> client: Running 'pre-boot' VM customizations...
==> client: Booting VM...
==> client: Waiting for machine to boot. This may take a few minutes...
      client: SSH address: 127.0.0.1:2222
      client: SSH username: vagrant
      client: SSH auth method: password
==> client: Machine booted and ready!
==> client: Checking for guest additions in VM...
      client: No guest additions were detected on the base box for this VM! Guest
      client: additions are required for forwarded ports, shared folders, host only
      client: networking, and more. If SSH fails on this machine, please install
      client: the guest additions and repackaging the box to continue.
      client:
      client: This is not an error message; everything may continue to work properly,
      client: in which case you may ignore this message.
==> client: Setting hostname...
```

- Обе VM успешно загружены в VirtualBox
- Выполнен вход под пользователем vagrant





- Подключение через `vagrant ssh server`
- Переход к пользователю `bahis`
- Аналогичное подключение к клиенту

```
C:\worker\bahis\vagrant>vagrant ssh server
VM must be running to open SSH connection. Run `vagrant up`
to start the virtual machine.
```

Рисунок 19: Подключение по SSH к server и переход к пользователю bahis


```
C:\worker\bahis\vagrant>vagrant ssh client
==> client: The machine you're attempting to SSH into is configured to use
==> client: password-based authentication. Vagrant can't script entering the
==> client: password for you. If you're prompted for a password, please enter
==> client: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password: █
```

Рисунок 20: Подключение по SSH к client и переход к пользователю bahis

- Выполнены `vagrant halt server` и `vagrant halt client`
- Корректное завершение работы ВМ

```
C:\worker\bahis\vagrant>vagrant halt server
==> server: Attempting graceful shutdown of VM...

C:\worker\bahis\vagrant>vagrant halt client
--> client: Attempting graceful shutdown of VM...
    client: Guest communication could not be established! This is usually because
    client: SSH is not running, the authentication information was changed,
    client: or some other networking issue. Vagrant will force halt, if
    client: capable.
==> client: Forcing shutdown of VM...
```

Рисунок 21: Остановка виртуальных машин server и client

2.3 Внесение изменений в настройки внутреннего окружения виртуальной машины

- В Vagrantfile добавлены блоки common user и common hostname
- Обеспечено выполнение 01-user.sh и 01-hostname.sh

```
# -*- mode: ruby -*-  
# vi: set ft=ruby :  
Vagrant.configure("2") do |config|  
  
  ## Common configuration  
  config.vm.provision "common dummy",  
  type: "shell",  
  preserve_order: true,  
  path: "provision/default/01-dummy.sh"  
  config.vm.provision "common user",  
  type: "shell",  
  preserve_order: true,  
  path: "provision/default/01-user.sh"
```

- Выполнены `vagrant up --provision` для `server` и `client`
- Повторный запуск provisioning
- Подтверждено существование пользователя `bahis`

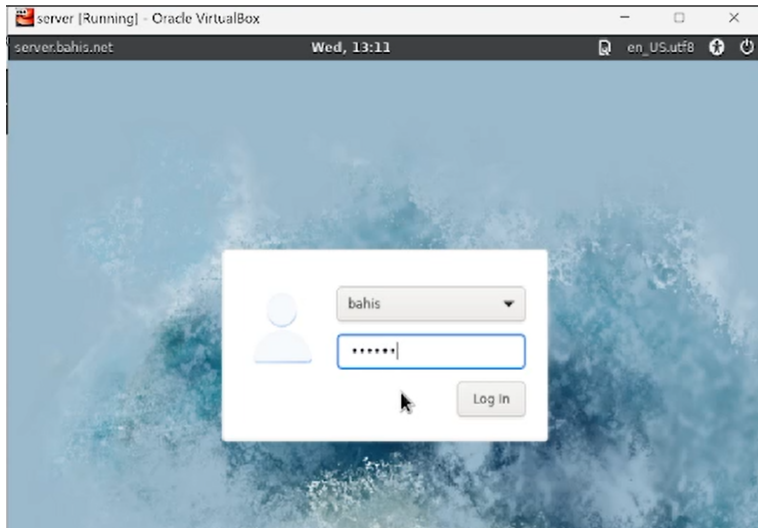
```
C:\worker\bahis\vagrant>vagrant up server --provision
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Clearing any previously set forwarded ports...
==> server: Clearing any previously set network interfaces...
==> server: Preparing network interfaces based on configuration...
server: Adapter 1: nat
server: Adapter 2: intnet
==> server: Forwarding ports...
server: 22 (guest) => 2222 (host) (adapter 1)
==> server: Running 'pre-boot' VM customizations...
==> server: Booting VM...
==> server: Waiting for machine to boot. This may take a few minutes...
server: SSH address: 127.0.0.1:2222
server: SSH username: vagrant
server: SSH auth method: password
==> server: Machine booted and ready!
==> server: Checking for guest additions in VM...
```

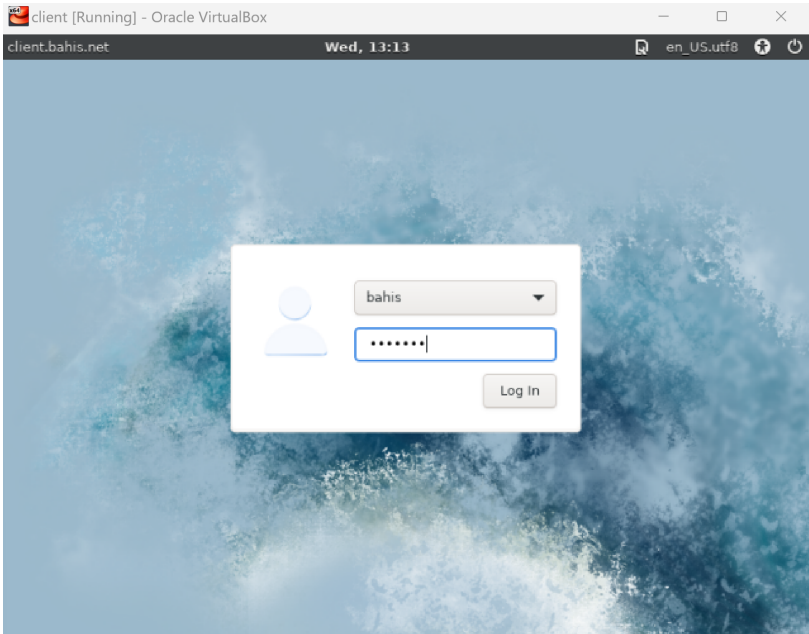
```
Administrator: Command Prompt - vagrant up client --provision
server: Provisioning script /tmp/vagrant-shell

C:\worker\bahis\vagrant>vagrant up client --provision
Bringing machine 'client' up with 'virtualbox' provider...
==> client: Clearing any previously set forwarded ports...
==> client: Fixed port collision for 22 => 2222. Now on port 2200.
==> client: Clearing any previously set network interfaces...
==> client: Preparing network interfaces based on configuration...
    client: Adapter 1: nat
    client: Adapter 2: intnet
==> client: Forwarding ports...
    client: 22 (guest) => 2200 (host) (adapter 1)
==> client: Running 'pre-boot' VM customizations...
==> client: Booting VM...
==> client: Waiting for machine to boot. This may take a few minutes...
    client: SSH address: 127.0.0.1:2200
    client: SSH username: vagrant
    client: SSH auth method: password
```

Рисунок 24: Повторный запуск client с ключом --provision

- Выполнен вход под пользователем bahis
- Проверено отображение FQDN в приглашении терминала
- Подтверждена корректная настройка hostname





Раздел 3

3. Выводы

3.1 Выводы

- Выполнена автоматическая сборка box-файла Rocky Linux через Packer
- Зарегистрирован образ в Vagrant
- Развёрнуты виртуальные машины server и client
- Применены provisioning-скрипты
- Настроены пользователь и hostname
- Подтверждена работа SSH и сетевых параметров
- Лабораторный стенд успешно подготовлен к дальнейшей работе